

Applied Machine Learning

Unit 04 — Lecture 06 Notes

Feature Engineering, Encoding and Dimensionality Reduction

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Read this first

These notes are written so that you can learn the material *without* having been in the lecture. Every concept is developed from scratch, every number you see was produced by running the code shown beside it, and every figure was generated from real computation rather than drawn by hand.

The previous lecture cleaned the table: holes filled, stuck sensors found, long tails straightened. It ended on one rule — *every number used to clean the data must be computed from the training rows only*. The table is now clean. It is not yet usable. The columns are in different units, some of them are words rather than numbers, and there may be far more of them than the problem needs.

Four questions organise this lecture. What should the columns **be**? What **units** should they be in? What do I do with a column of **words**? And what do I do when there are **too many columns**?

The single result to carry out of the lecture: a logistic regression scored **0.8475** on a table of 500 pure-noise features and 60 rows whose labels were coin flips. The only thing wrong with the experiment was that the features were chosen before the data was split. Work through Section 7.8 with a Python session open until you can say exactly why.

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1 How to use these notes

Read in order. Section 6 assumes the scaling material of Section 3, and Section 7 contrasts itself throughout with Section 6. Everything here assumes only what Lecture 5 assumed: a train/test split, a fitted model and a score. The one new piece of mathematics is the eigen-decomposition of a 2×2 symmetric matrix, done by hand in Section 6.2.

There are four kinds of box. **Key idea** — the one sentence to remember a month from now. **Worked example** — a calculation done by hand, with small numbers, before any library is

used. **Caution** — the specific mistake that section exists to prevent. **Try it yourself** — something to run, with the expected output given so you can check.

Code appears in a framed block, and directly beneath it a shaded block showing what that code *actually printed*. If you run it and get something different, trust your machine and tell me.

1.1 What you need

```
pip install numpy scipy scikit-learn matplotlib pandas
```

Every number in these notes was produced with:

```
import numpy, scipy, sklearn, matplotlib
print("numpy", numpy.__version__, "| scipy", scipy.__version__,
      "| sklearn", sklearn.__version__,
      "| matplotlib", matplotlib.__version__)
```

```
numpy 2.2.6 | scipy 1.15.3 | sklearn 1.7.2 | matplotlib 3.10.9
```

Four bundled datasets recur, and **none of them downloads anything**.

```
from sklearn.datasets import (load_breast_cancer, load_digits,
                              load_wine, load_diabetes)
bc = load_breast_cancer(); Xb, yb = bc.data, bc.target
dg = load_digits(); Xg, yg = dg.data, dg.target
wine = load_wine(); Xw, yw = wine.data, wine.target
dia = load_diabetes(); Xd, yd = dia.data, dia.target.astype(float)
```

```
load_breast_cancer 569 rows x 30 features, 2 classes
load_digits        1797 rows x 64 features, 10 classes
load_wine          178 rows x 13 features, 3 classes
load_diabetes       442 rows x 10 features, continuous target
missing cells anywhere: 0
feature scales in breast_cancer differ by a factor of 215171
```

That last line is why `load_breast_cancer` is the running example: its widest column has a standard deviation two hundred thousand times the narrowest, which is exactly the situation this lecture is about.

Four datasets, one seed — the convention for this lecture

Everything in these notes is computed from the four bundled tables above, plus a handful of small problems drawn on the spot to make a specific point, and **one seed**. The seed is **0**: every generator is created as `np.random.default_rng(0)`, every stochastic estimator is given `random_state=0` — including `mutual_info_classif`, which is stochastic and silently gives you a different answer without one — and the cross-validation shuffle is `StratifiedKFold(5, shuffle=True, random_state=0)`. Every listing below that involves chance shows that line.

The exceptions are the two *sweeps*: the forty train/test splits of Section 6.7 and the twenty repetitions of Section 7.8 use a different seed each time, on purpose, because the variation between repetitions is the thing being measured. Their seeds are still derived from the same 0, so the sweep as a whole reproduces exactly.

Run the code as written and you get the numbers printed here. If you get something different, the difference is in your code or your library versions, not in chance.

Do not reach for `fetch_*`

Every `fetch_*` loader downloads over HTTPS, and on the campus network those downloads fail with a certificate error you cannot fix from your machine. Nothing in this course depends on one: use the bundled loaders (`load_diabetes`, `load_breast_cancer`, `load_iris`, `load_wine`, `load_digits`) or a fixed-seed synthetic set.

1.2 Learning outcomes

By the end of this lecture you should be able to:

1. Construct new features by combining, decomposing, binning and transforming existing ones, and say which models such engineering helps and which it does not (CO2).
2. Apply standardisation, min-max, robust and max-abs scaling; derive each from its formula on a small vector by hand; and predict which learning algorithms change their answer when the data is rescaled and which cannot (CO2).
3. Encode nominal and ordinal categorical variables correctly, explain and avoid the dummy variable trap, and quantify the cost of one-hot encoding a high-cardinality column (CO2).
4. State Han, Kamber and Pei's three kinds of data reduction — dimensionality, numerosity and compression — and give a worked example of each (CO2).
5. Compute a principal component analysis by hand on a small dataset, interpret the variance-explained curve, choose k , quantify reconstruction error, and explain why PCA must be fitted on the training split and preceded by scaling (CO2).
6. Distinguish filter, wrapper and embedded feature selection; apply variance thresholds, ANOVA F , mutual information, recursive feature elimination and L1 regularisation; and identify selection bias in a described workflow (CO2).

2 Feature engineering

2.1 What it is, and why it is worth more than the algorithm

Feature engineering is choosing the representation of the data that the model sees. It is the least glamorous and most consequential thing in the subject, and Müller and Guido open their chapter on it with the claim that *how you represent your data matters more than which algorithm you pick*.

Four moves cover almost everything you will do.

1. **Combine.** Length and width become area. Height and weight become body mass index. Two timestamps become a duration.
2. **Decompose.** A timestamp becomes hour-of-day, day-of-week and month. An address becomes a postcode and a city.
3. **Bin.** A continuous variable becomes a set of intervals, so a linear model can fit a step function.
4. **Transform.** log, square root, ratios, differences. This is the Lecture 5 material seen from the other side.

Key idea

A linear model can only add its inputs up. If the quantity that matters is a product, a ratio or a threshold, no amount of optimisation will find it — but one new column will.

2.2 Two measures used throughout: correlation and R^2

Almost every experiment in this lecture is read through one of two numbers. Both are defined here once, so that the rest of the notes can simply quote them.

Correlation. The (Pearson) correlation of two columns x and y over n rows is

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}, \quad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \quad S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2. \quad (1)$$

It always lies in $[-1, 1]$. It is $+1$ when the points lie exactly on a rising straight line, -1 on a falling one, and near 0 when there is no *straight-line* tendency. It does not change if you rescale or shift either column, which is why it is written without units. In NumPy it is `np.corrcoef(x, y)[0, 1]`; in pandas, `df.corr()`.

The coefficient of determination, R^2 . For a model that predicts $\hat{y}_i$,

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}, \quad SS_{\text{res}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad SS_{\text{tot}} = \sum_{i=1}^n (y_i - \bar{y})^2. \quad (2)$$

SS_{tot} is the error of the laziest possible model, one that always predicts the mean. So $R^2 = 1$ is a perfect fit, $R^2 = 0$ is no better than the mean, and $R^2 < 0$ — which Lecture 5 met with listwise deletion — is *worse* than the mean. On held-out data it can be as negative as the model is bad; there is no lower limit. Every scikit-learn regressor's `.score(X, y)` returns it, as do `sklearn.metrics.r2_score(y, y_pred)` and `cross_val_score(..., scoring="r2")`.

How the two are related. For a straight line fitted by least squares on a *single* input, $R^2 = r^2$. With several inputs there is no single r , and R^2 is the number to use.

Correlation only sees straight lines

$r \approx 0$ does not mean “unrelated”. For $y = x^2$ on x symmetric about zero, $r = 0$ exactly while y is a perfect function of x . The next example is the same trap in a milder form.

2.3 Worked: four plots of land

The information is there; the representation is hiding it

Four rectangular plots. The price is 5 per square metre, in rupees. Nobody recorded the area.

length ℓ (m)	10	20	30	40
width w (m)	40	15	20	12
area = ℓw	400	300	600	480
price = $5\ell w$	2000	1500	3000	2400

The price is a perfect deterministic function of the two recorded columns, but it is not a *linear* one, and the correlations show it:

$$\text{corr}(\text{price}, \ell) = +0.5494, \quad \text{corr}(\text{price}, w) = -0.0948, \quad \text{corr}(\text{price}, \ell w) = +1.0000.$$

For the length column, Equation (1) by hand: $\bar{\ell} = 25$ and $\overline{\text{price}} = 2225$, so the deviations are $(-15, -5, 5, 15)$ and $(-225, -725, 775, 175)$, giving

$$S_{xy} = 13\,500, \quad S_{xx} = 500, \quad S_{yy} = 1\,207\,500, \quad r = \frac{13\,500}{\sqrt{500 \times 1\,207\,500}} = 0.5494.$$

A straight line on length alone would therefore have $R^2 = 0.5494^2 = 0.3019$.

The width even appears *negatively* associated with price, which is an artefact of the four plots we happen to have. A linear model on (ℓ, w) leaves $\text{SS}_{\text{res}} = 415\,604$ of the $\text{SS}_{\text{tot}} = 1\,207\,500$, so by Equation (2)

$$R^2 = 1 - \frac{415\,604}{1\,207\,500} = 0.6558;$$

add the column ℓw and SS_{res} falls to zero, so $R^2 = 1.0000$.

```
length = np.array([10., 20., 30., 40.])
width = np.array([40., 15., 20., 12.])
price = 5.0 * length * width
np.corrcoef(price, length)[0, 1] # likewise width, length * width
LinearRegression().fit(np.c_[length, width], price).score(...)
LinearRegression().fit(np.c_[length, width, length * width], price).score(...)
```

```
corr(price, length) = +0.5494
corr(price, width ) = -0.0948
corr(price, area  ) = +1.0000
R2 of a linear model on (length, width)      0.6558
R2 of a linear model on (length, width, area) 1.0000
```

Four rows is too few to be convincing. Here is the same idea on 400 rows with noise, cross-validated.

```
rng = np.random.default_rng(0) # the one seed for this lecture
x1, x2 = rng.uniform(1, 10, 400), rng.uniform(1, 10, 400)
y = 3.0 * x1 * x2 + rng.normal(0, 3.0, 400)
for X in (np.c_[x1, x2], np.c_[x1, x2, x1 * x2]):
    print(cross_val_score(LinearRegression(), X, y, cv=5, scoring="r2").mean())
```

```
linear model on x1, x2      cross-validated R2 0.9091
linear model on x1, x2, x1*x2  cross-validated R2 0.9981
a random forest on the raw pair gets there too, accuracy 0.9650
```

Read the third line as carefully as the first two. A tree finds an interaction unaided, by splitting twice; the engineered column is worth a great deal to a linear model and almost nothing to a forest. **Feature engineering is worth most to the simplest models**, which is one reason simple models plus good features remain competitive.

2.4 Binning, and why a representation is only good relative to a model

`KBinsDiscretizer` cuts a continuous column into intervals and one-hot encodes the interval. A linear model on those indicator columns is a step function, so binning buys a linear model the flexibility it did not have.

```
kb = KBinsDiscretizer(n_bins=10, encode="onehot-dense", strategy="uniform")
cross_val_score(make_pipeline(kb, LinearRegression()), x, y, cv=5, scoring="r2")
```

```
one wiggly feature, 250 points, y = sin(4x) + x + noise
               raw x      10 bins
LinearRegression      0.8194      0.9151
DecisionTreeRegressor 0.9475      0.9151
```

```
binning bought the linear model +0.0957 and cost the tree -0.0323
```

The two scores after binning are *identical to four decimal places*, and that is not a coincidence: after binning, both models are fitting a step function on the same ten intervals, so they are fitting the same family. The tree lost 0.0323 because it had been choosing its own cut points and we took that away.

Key idea

A representation is not good or bad in the abstract. It is good or bad *relative to a model*. Binning helps linear models and hurts trees.

2.5 Engineering blind: the combinatorial blow-up

If products help, why not add all of them? `PolynomialFeatures(degree=k)` does exactly that, and the count is $\binom{d+k}{k} - 1$, which grows like d^k .

d=2	degree 2 ->	5 features	d=2	degree 3 ->	9 features
d=10	degree 2 ->	65 features	d=10	degree 3 ->	285 features
d=30	degree 2 ->	495 features	d=30	degree 3 ->	5455 features
d=64	degree 2 ->	2144 features	d=64	degree 3 ->	47904 features

```
X2 = PolynomialFeatures(degree=2, include_bias=False).fit_transform(Xd)
cross_val_score(make_pipeline(StandardScaler(), Ridge(1.0)), X2, yd,
                 cv=5, scoring="r2").mean()
```

diabetes, ridge on 10 raw features	R2 0.4822
diabetes, ridge on 65 degree-2 features	R2 0.4179
difference	-0.0643

Caution

On 442 patients, going from 10 columns to 65 cost six points of R^2 even with ridge regularisation, and not one of the 55 new products means anything clinically. **Engineer a feature because you have a reason, not because a function will generate one for you.**

Try it yourself

Run `PolynomialFeatures(degree=2).fit(np.zeros((2,d)))` for $d = 5, 20, 100$ and check the count against $\binom{d+2}{2} - 1$. For $d = 100$ you should get 5150.

Sources: Han, Kamber and Pei, *Data Mining: Concepts and Techniques* (3e), Ch. 3.; Müller and Guido, *Introduction to Machine Learning with Python*, Ch. 3–4.; James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*, Ch. 5–6, 12.; Scikit-learn documentation (preprocessing, model selection), accessed 2026-08-04.

3 Feature scaling

3.1 Three mechanisms by which units matter

1. **Distances.** k -nearest neighbours, support vector machines with an RBF kernel, and k -means all add squared differences across columns. A column measured in large numbers contributes large squares and dominates the sum.
2. **Gradients.** Unequal scales make the loss surface a long thin ravine. Gradient descent must use a step size small enough for the steepest direction, which makes it crawl along the shallow one.

3. **Penalties.** Ridge and lasso shrink $\|w\|$. If w_1 is in rupees per square metre and w_2 in rupees per year, adding w_1^2 to w_2^2 is adding two things that are not the same kind of thing.

And one family that genuinely does not care: **trees**. A split asks “is $x_j \leq t$?”. Any increasing rescaling $x \mapsto (x - a)/b$ with $b > 0$ moves t to $(t - a)/b$ and sends exactly the same rows left.

Caution

Scaling is *not* the transformation of Lecture 5. Box–Cox changes the **shape** of a distribution; scaling changes only its **location and width**. Skewness and kurtosis are unchanged by every scaler in this section. If a column was lopsided before, it is lopsided afterwards.

3.2 Worked: three customers

Where a squared distance actually comes from

	age (years)	annual income (in rupees)
A	25	400,000
B	60	405,000
C	26	800,000

Squared Euclidean distance is a sum of per-column contributions:

$$d(A, B)^2 = \underbrace{35^2}_{1,225} + \underbrace{5000^2}_{25,000,000} = 25,001,225, \quad d(A, C)^2 = \underbrace{1^2}_1 + \underbrace{400,000^2}_{1.6 \times 10^{11}}.$$

Income supplies 99.995% of the first and 100.000% of the second. The age column is in the file, in the formula, and absent from the answer.

Now standardise each column with $z = (x - \bar{x})/\sigma$ (population σ , which is what **StandardScaler** uses). For age, $\bar{x} = 37$ and $\sigma = 16.269$; for income, $\bar{x} = 535,000$ and $\sigma = 187,395$.

	z_{age}	z_{income}
A	−0.7376	−0.7204
B	+1.4138	−0.6937
C	−0.6762	+1.4141

$$d(A, B)^2 = 4.6285 + 0.0007 = 4.6292, \quad d(A, C)^2 = 0.0038 + 4.5562 = 4.5600.$$

The nearest neighbour of A has changed from B to C. Age now supplies 99.98% of the first distance and income 99.92% of the second, which is the honest description of the two comparisons: A and B differ mostly in age; A and C differ mostly in income.

Key idea

Choosing not to scale is not neutral. It is choosing to weight each feature by its standard deviation, which is a decision about importance that you did not make deliberately and cannot defend.

3.3 Four scalers

	Formula	Output	Broken by an outlier?
StandardScaler	$\frac{x - \bar{x}}{\sigma}$	mean 0, sd 1	yes: both $\bar{x}$ and σ move
MinMaxScaler	$\frac{x - x_{\min}}{x_{\max} - x_{\min}}$	exactly $[0, 1]$	completely : it is defined by the extremes
RobustScaler	$\frac{x - \tilde{x}}{\text{IQR}}$	median 0, IQR 1	no: a median and two quartiles
MaxAbsScaler	$\frac{x}{\max x }$	$[-1, 1]$, zeros preserved	yes, but it is sparse-safe

Every one of these is of the form $x \mapsto (x - a)/b$. The differences are entirely in how a and b are estimated.

Eight numbers through four scalers

$v = 2, 4, 4, 4, 5, 5, 7, 9$.

Mean 5, population sd 2, min 2, max 9, median 4.5, $Q_1 = 4$, $Q_3 = 5.5$, IQR = 1.5. Check the first entry of each row: $(2 - 5)/2 = -1.5$; $(2 - 2)/(9 - 2) = 0$; $(2 - 4.5)/1.5 = -1.6667$; $2/9 = 0.2222$.

```
V = np.array([2., 4., 4., 4., 5., 5., 7., 9.]).reshape(-1, 1)
for sc in (StandardScaler(), MinMaxScaler(), RobustScaler(), MaxAbsScaler()):
    print(sc.fit_transform(V).ravel())
```

```
mean 5.0000 sd(ddof=0) 2.0000 min 2.0 max 9.0
median 4.5000 Q1 4.0000 Q3 5.5000 IQR 1.5000
StandardScaler [-1.5 -0.5 -0.5 -0.5  0.  0.  1.  2. ]
MinMaxScaler [0.  0.2857 0.2857 0.2857 0.4286 0.4286 0.7143 1.  ]
RobustScaler [-1.6667 -0.3333 -0.3333 -0.3333  0.3333  0.3333  1.6667  3.  ]
MaxAbsScaler [0.2222 0.4444 0.4444 0.4444 0.5556 0.5556 0.7778 1.  ]
```

Note that **RobustScaler** does not bound its output: the value 9 lands at 3.0. That is deliberate — it refuses to compress the data to make room for its extremes.

3.4 One typo, and which scaler survives it

Append a single mistyped 500 to the same eight numbers.

```
StandardScaler the 8 good values now span 0.0450 (was 3.5000)
MinMaxScaler  the 8 good values now span 0.0141 (was 1.0000)
RobustScaler   the 8 good values now span 2.3333 (was 4.6667)

compression factor suffered by the 8 good values
StandardScaler 77.8 times narrower
MinMaxScaler   71.1 times narrower
RobustScaler    2.0 times narrower
why: the centre and the width each scaler uses
mean  5.0000 -> 60.0000 sd  2.0000 -> 155.5749
min   2.0000 ->  2.0000 max  9.0000 -> 500.0000
median 4.5000 ->  5.0000 IQR  1.5000 ->  3.0000
```

Look at the last three lines, because they explain everything above them. The mean moved by a factor of 12 and the standard deviation by a factor of 78. The maximum moved by a factor of 56. **The median moved from 4.5 to 5.0 and the interquartile range from 1.5 to 3.0** — and that factor of two is the entire damage **RobustScaler** sustained.

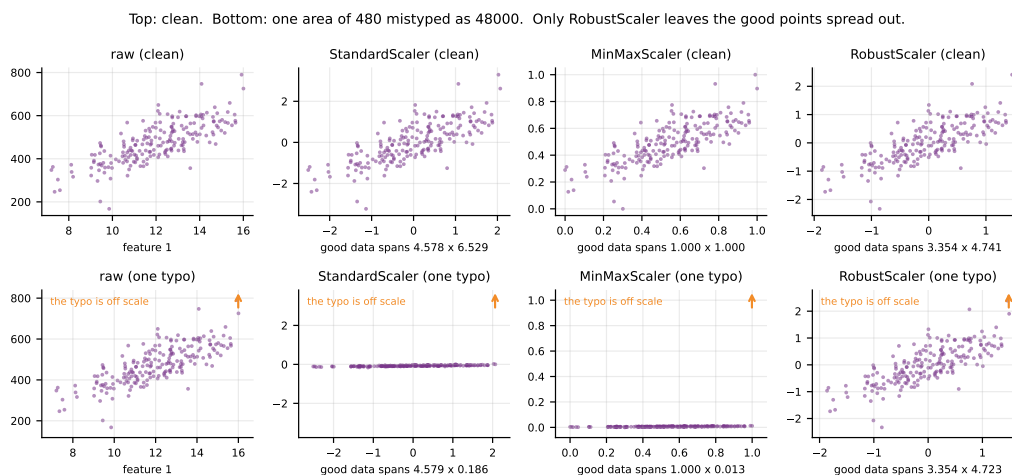


Figure 1: Two hundred plots of land, with length on the horizontal axis and area on the vertical. Top row: clean. Bottom row: one area of 480 typed as 48000. The bottom frames are the *same* frames as the top ones, so the collapse is visible; the typo itself is off scale. Under **MinMaxScaler** the good data ends up occupying 1.3% of the unit interval.

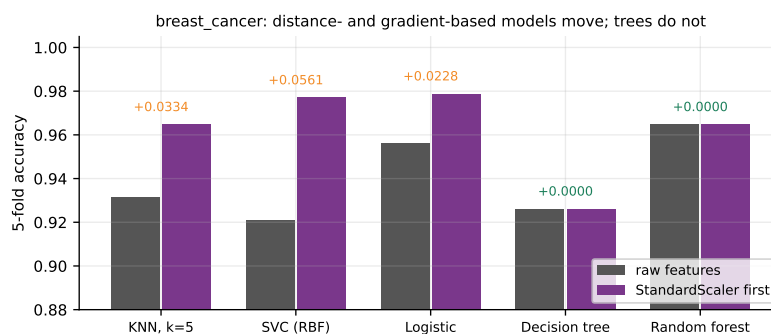


Figure 2: The same table drawn. The three left-hand pairs are models that measure a distance or descend a gradient; the two right-hand pairs are models that only ever ask which side of a threshold a value falls on.

Caution

MinMaxScaler is the dangerous one, because its output is *guaranteed* to be in $[0,1]$ and therefore looks correct. Nothing in the output tells you that 99% of your rows are inside a window of width 0.013. If you have not run an outlier check (Lecture 5), do not use **MinMaxScaler**.

3.5 Which models change, measured

```
cv = StratifiedKFold(5, shuffle=True, random_state=0)
for name, m in models:
    print(cross_val_score(m, Xb, yb, cv=cv).mean(),
          cross_val_score(make_pipeline(StandardScaler()), m), Xb, yb, cv=cv).mean())
```

model	raw	scaled	change
KNeighborsClassifier(5)	0.9315	0.9649	+0.0334
SVC(rbf)	0.9210	0.9771	+0.0561
LogisticRegression	0.9543	0.9789	+0.0246
DecisionTreeClassifier	0.9262	0.9262	+0.0000
RandomForestClassifier	0.9649	0.9649	+0.0000

Those two zeros are not roundings. Here is the proof.

```
t1 = DecisionTreeClassifier(random_state=0).fit(X_train, y_train)
sc = StandardScaler().fit(X_train)
```

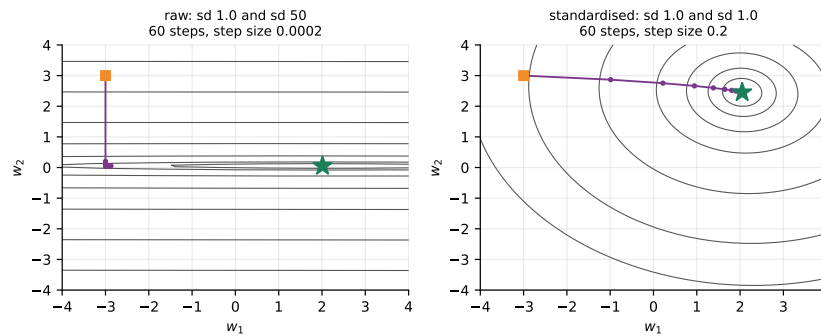


Figure 3: Two features, one with fifty times the spread of the other. Orange square is the start, green star the least-squares solution, purple the gradient-descent path. Left: Hessian condition number 2296.5; after 60 steps the loss is 323 times the optimum. Right: condition number 1.1, and 60 steps arrive.

```
t2 = DecisionTreeClassifier(random_state=0).fit(sc.transform(X_train), y_train)
np.array_equal(t1.tree_.feature, t2.tree_.feature)
```

```
tree on raw      : 18 leaves, depth 5, test accuracy 0.902098
tree on scaled  : 18 leaves, depth 5, test accuracy 0.902098
identical predictions on the test set: True
identical tree structure (same feature at every node): True
the thresholds differ, of course: raw 106.1000  scaled -0.0273 (same split)
```

Key idea

A tree is invariant under any strictly increasing per-column transformation, because every question it asks is about *order*. Scaling a tree's inputs is not wrong; it is simply a no-op that costs you the ability to read the thresholds.

3.6 The gradient case

```
for eta in (1e-2, 1e-3, 1e-4, 1e-5, 1e-6):
    for X in (X_raw, StandardScaler().fit_transform(X_raw)):
        m = SGDRegressor(max_iter=50000, tol=1e-6, random_state=0,
                          learning_rate="constant", eta0=eta).fit(X, y)
        print(eta, m.n_iter_, m.score(X, y))
```

eta0		raw units	standardised
0.01	6 epochs	R2 -1.7e+28	15 epochs R2 +0.4951
0.001	6 epochs	R2 -4.8e+26	14 epochs R2 +0.5142
0.0001	8 epochs	R2 -4.2e+25	184 epochs R2 +0.5144
1e-05	24 epochs	R2 +0.2008	872 epochs R2 +0.5139
1e-06	42 epochs	R2 +0.3478	5231 epochs R2 +0.5113

ordinary least squares on the same data gives R2 +0.5177
condition number of $X^T X$, raw 1.030e+06
condition number of $X^T X$, standardised 4.701e+02

There is no step size at which the unscaled version works. Large steps diverge — an R^2 of -1.7×10^{28} is an overflow, not a bad fit — and small steps stop early at 0.35, because the tolerance is met while the shallow direction has barely moved. Standardised, *every* step size lands within 0.007 of least squares, the best in fourteen passes.

The condition number — the ratio of the largest to the smallest eigenvalue of $X^T X$ — is the quantitative version of that picture, and it bounds how slowly gradient descent can converge. Standardising took it from a million to four hundred and seventy.

3.7 And the scaler is fitted on the training split

Everything Lecture 5 said applies unchanged. `StandardScaler` learns two numbers per column, `MinMaxScaler` learns two, `RobustScaler` learns two. All of them must come from the training rows.

```
# wrong
X_all = StandardScaler().fit_transform(X)
X_tr, X_te, y_tr, y_te = train_test_split(X_all, y, random_state=0)

# right
pipe = make_pipeline(StandardScaler(), KNeighborsClassifier())
cross_val_score(pipe, X, y, cv=5)
```

Lecture 5 measured the size of that leak: +0.0034 of accuracy with 20 training rows, falling to +0.0002 with 427. Real, and small. The much larger leak of this lecture is in Section 7.8.

Sources: Han, Kamber and Pei, *Data Mining: Concepts and Techniques* (3e), Ch. 3.; Müller and Guido, *Introduction to Machine Learning with Python*, Ch. 3–4.; James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*, Ch. 5–6, 12.; Scikit-learn documentation (preprocessing, model selection), accessed 2026-08-04.

4 Feature encoding

4.1 Nominal and ordinal

A categorical column is **ordinal** if its categories have a genuine order — `small < medium < large`, a degree classification, a Likert response. It is **nominal** if they do not — colour, city, blood group, department, payment method.

There are two standard encodings. **Ordinal encoding** replaces each category with an integer, giving one column. **One-hot encoding** creates one 0/1 column per category, with exactly one 1 in each row.

Key idea

Match the encoding to the variable, not to your convenience. Ordinal-encoding a nominal variable asserts an order and a spacing that are not there, and a linear model believes assertions.

4.2 Worked: what a linear model does with an ordinal code

Three colours

`OrdinalEncoder` sorts alphabetically, so `blue` → 0, `green` → 1, `red` → 2. Suppose the true mean target for each colour is 20, 50 and 10.

A linear model on the code column fits the best straight line through the three points (0, 20), (1, 50), (2, 10). With $\bar{x} = 1$ and $\bar{y} = 80/3 = 26.667$:

$$S_{xy} = (-1)(-6.667) + (0)(23.333) + (1)(-16.667) = -10, \quad S_{xx} = 1 + 0 + 1 = 2,$$

so the slope is -5 and the intercept is $26.667 + 5 = 31.667$. The three predictions are 31.667, 26.667, 21.667 against truths of 20, 50, 10.

$$\text{SSE} = 11.667^2 + 23.333^2 + 11.667^2 = 816.7, \quad \text{SST} = 6.667^2 + 23.333^2 + 16.667^2 = 866.7,$$

$$R^2 = 1 - \frac{816.7}{866.7} = 0.0577.$$

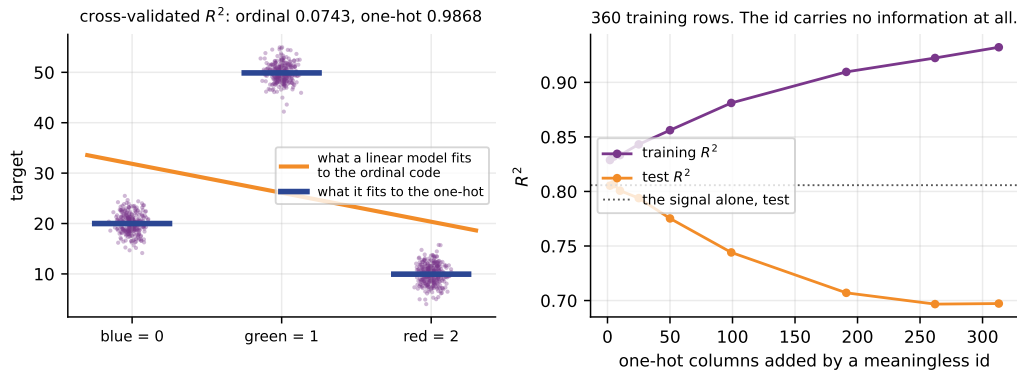


Figure 4: Left: the orange line is the whole vocabulary a linear model has when given an ordinal code; the three blue bars are what one-hot lets it say. Right: the price of the opposite mistake, one-hot encoding an identifier that carries no information.

The point is structural, not numerical. A straight line through codes 0, 1, 2 *cannot* put the middle category highest, whatever its slope. Ordinal coding has thrown away the ability to express the pattern.

```
rng = np.random.default_rng(0) # the one seed for this lecture
colour = rng.choice(["blue", "green", "red"], size=900)
y = np.array([{"blue":20., "green":50., "red":10.}[c] for c in colour]) \
    + rng.normal(0, 2.0, 900)
ordc = OrdinalEncoder().fit_transform(colour.reshape(-1, 1))
ohc = OneHotEncoder(sparse_output=False).fit_transform(colour.reshape(-1, 1))
```

```
linear model, ordinal (1 column)    cross-validated R2 +0.0743
linear model, one-hot (3 columns)    cross-validated R2 +0.9868
a decision tree on the SAME ordinal column: accuracy 1.0000
```

0.0743 against 0.9868, from a choice of encoder. The hand calculation predicted 0.0577; the difference is the noise we added. And once again the third line matters: a tree on the ordinal column recovers the pattern perfectly, because two splits on a code column separate three groups. **The failure is a failure of the linear model plus that encoding, not of the encoding alone.**

4.3 The dummy variable trap

If a design matrix has an intercept column of ones *and* all k dummy columns, then the k dummies sum to 1 in every row, which is exactly the intercept column. The matrix is rank-deficient and the coefficients are not unique.

```
D_full = np.c_[np.ones(900), OneHotEncoder(sparse_output=False).fit_transform(C)]
D_drop = np.c_[np.ones(900), OneHotEncoder(sparse_output=False,
                                             drop="first").fit_transform(C)]
print(np.linalg.matrix_rank(D_full), np.linalg.cond(D_full))
```

```
with an intercept and all 3 dummies: shape (900, 4) rank 3
    the three dummy columns sum to [1.] in every row
    condition number 3.436e+15
with an intercept and 2 dummies:      shape (900, 3) rank 3
    condition number 3.810e+00
coefficients, all 3 dummies : [ 19.9581  0.0382 29.926 -10.0061]
a DIFFERENT coefficient vector: [ 119.9581 -99.9618 -70.074 -110.0061]
    same predictions? max difference 8.88e-15
both fit equally well: R2 0.986928 and 0.986928
```

The second coefficient vector was produced by adding 100 to the intercept and subtracting 100

from each dummy. The predictions move by at most 9×10^{-15} — floating-point dust on values near 20 — because the change is $100 \times (1 - \sum_j \text{dummy}_j) = 0$ in every row.

Caution

Rank deficiency does not make your *predictions* wrong — both fits give $R^2 = 0.986928$. It makes your *coefficients* meaningless, and with them every standard error, *t*-statistic and *p*-value. If anyone is going to interpret the model, use `drop='first'`. Ridge, lasso and trees are unaffected; ordinary least squares and statistical inference are not.

Two practical notes: with ridge or lasso, dropping a level makes the penalty asymmetric across the retained levels, so many practitioners keep all k ; and `drop='if_binary'` is usually what you want for a mixed table.

4.4 High cardinality

One-hot encoding a column with K levels adds K columns. That is fine for $K = 3$ and catastrophic for $K = 3000$.

```
the id column has 256 distinct levels in 600 rows; one-hot makes 256 columns
signal only          train R2 +0.7586   test R2 +0.8063
signal + one-hot id  train R2 +0.8956   test R2 +0.6825
levels present in train but not test: 99
```

The identifier was generated at random and carries *no* information about the target. Adding it moved the training R^2 up by 0.14 and the test R^2 down by 0.12. Each dummy column fits a handful of rows exactly and predicts nothing, which is a textbook picture of overfitting. And 82 of the levels never appear in the test set at all, so their coefficients were estimated from data that will never recur.

What to do instead: group rare levels into a single **other**; use a count or target encoding, fitted *inside* the cross-validation fold like any other learned quantity; or use a model with native categorical support.

4.5 Two details that will bite you

```
enc = OneHotEncoder().fit(["small", "medium", "large", "small"])
enc.transform(["extra-large"])
```

```
OneHotEncoder() default -> ValueError: Found unknown categories
      [np.str_('extra-large')] in column 0 during transform
handle_unknown='ignore' -> all-zero row:
[[0. 0. 1.]
 [0. 0. 0.]
```

An unseen category at prediction time raises by default. `handle_unknown = 'ignore'` produces an all-zero row instead, which is a defensible choice provided you know you have made it.

```
OrdinalEncoder().fit_transform(sizes) # ALPHABETICAL
OrdinalEncoder(categories=["small", "medium", "large"]) # what you meant
```

```
alphabetical (the default): [2. 1. 0. 1. 2.]
      categories_ = ['large' 'medium' 'small']
stated order              : [0. 1. 2. 1. 0.]
```

Caution

The default ordering put **large** before **medium**. If your variable really is ordinal you must pass `categories=` explicitly; the encoder cannot read English. This is a silent bug —

nothing raises, the model fits, and the number it reports is simply worse than it should be.

Sources: Han, Kamber and Pei, *Data Mining: Concepts and Techniques* (3e), Ch. 3.;Müller and Guido, *Introduction to Machine Learning with Python*, Ch. 3–4.;James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*, Ch. 5–6, 12.;Scikit-learn documentation (preprocessing, model selection), accessed 2026-08-04.

5 Data reduction concepts

5.1 Han, Kamber and Pei’s three kinds

	What it removes	Methods
Dimensionality reduction	columns	attribute subset selection (Section 7), principal component analysis (Section 6), wavelet transforms
Numerosity reduction	rows, or the raw values	<i>parametric</i> : fit a model and keep its parameters instead of the data; <i>non-parametric</i> : histograms, clustering, sampling
Data compression	bits	lossless (run-length and entropy coding); lossy (PCA and wavelets again, seen from the other side)

The test that all three must pass is the same: **the reduced dataset must support the same analysis and give substantially the same answer**, and the reduction must cost less than the saving it produces.

5.2 Why fewer columns: the curse of dimensionality

```
rng = np.random.default_rng(0) # the one seed for this lecture
Q = rng.random((500, d))
D = np.sqrt(((Q[:, None, :] - Q[None, :, :]) ** 2).sum(-1))
np.fill_diagonal(D, np.inf); near = D.min(axis=1)
np.fill_diagonal(D, -np.inf); far = D.max(axis=1)
print(d, near.mean(), far.mean(), np.mean(far / near))
```

	d	mean	dnear	mean	dfar	dfar/dnear
1		0.0010		0.7488		4425.9149
2		0.0232		1.0332		67.5754
5		0.2117		1.4525		7.3998
10		0.5495		1.8870		3.5278
50		2.1608		3.5227		1.6329
100		3.3785		4.7283		1.4004
500		8.4293		9.7936		1.1620
1000		12.2254		13.6046		1.1129

At $d = 1000$ the farthest of the 499 other points is only 1.11 times as far away as the nearest. Every pair of points is at very nearly the same distance, so “nearest neighbour” has stopped carrying information.

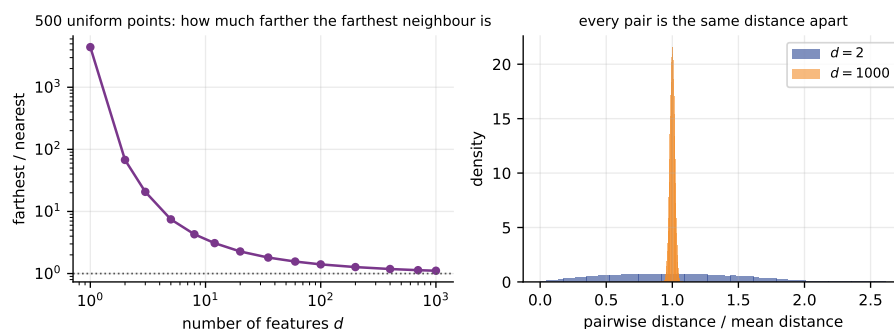


Figure 5: Left: the farthest-to-nearest ratio against d , log-log. Right: the whole distribution of pairwise distances at $d = 2$ and $d = 1000$, each divided by its own mean. At $d = 1000$ every distance in the dataset is within a few per cent of every other.

Key idea

Every method that measures a distance — k -NN, k -means, RBF kernels, DBSCAN — degrades as the number of columns grows, and it degrades quietly. Nothing in the output says “the distances are meaningless now”.

5.3 Numerosity reduction: keep fewer rows

digits, all	1797 rows	accuracy	0.9694	
digits, a	50% sample (898 rows)	accuracy	0.9607	loss -0.0087
digits, a	25% sample (449 rows)	accuracy	0.9474	loss -0.0220
digits, a	10% sample (179 rows)	accuracy	0.9134	loss -0.0560
digits, a	5% sample (89 rows)	accuracy	0.8506	loss -0.1188

Half the data costs under one point of accuracy; a twentieth costs twelve. That shape — flat at the top, steep at the bottom — is the learning curve, and it is the argument for the workflow you should adopt now: **prototype on a sample, fit on everything**. It is also how you make a 200-repetition experiment finish before the lab period ends.

Sources: Han, Kamber and Pei, *Data Mining: Concepts and Techniques* (3e), Ch. 3.; Müller and Guido, *Introduction to Machine Learning with Python*, Ch. 3–4.; James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*, Ch. 5–6, 12.; Scikit-learn documentation (preprocessing, model selection), accessed 2026-08-04.

6 Dimensionality reduction: principal component analysis

6.1 The question PCA asks

Of all the directions in feature space, which one do the data vary along the most? Then, among directions perpendicular to that one, which is next? And so on. The recipe:

1. Centre every column by subtracting its mean, giving X_c .
2. Form the covariance matrix $S = \frac{1}{n-1} X_c^T X_c$.
3. Compute the eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ and their eigenvectors $u_1, u_2, \dots$ of S .
4. Keep the top k . The **scores** are $T = X_c U_k$.

Two facts make the whole thing readable. First, λ_j is the variance of the data along component j , so $\lambda_j / \sum_i \lambda_i$ is the **fraction of variance explained**. Second, $\sum_i \lambda_i = \text{tr}(S)$, the total variance of the original columns — a rotation moves variance around but never creates or destroys it.

There is a second, equivalent characterisation that is often more useful: **the k -dimensional subspace PCA finds is the one that minimises the total squared reconstruction error**. Maximising retained variance and minimising discarded variance are the same problem.

6.2 Worked: five points, three features

PCA with no library at all

Take the five rows

$$(1, 2, 5), \quad (2, 3, 4), \quad (3, 5, 6), \quad (4, 4, 6), \quad (5, 6, 4),$$

so each observation now has three features rather than two. The mean is $(3, 4, 5)$ and the centred rows are

$$(-2, -2, 0), \quad (-1, -1, -1), \quad (0, 1, 1), \quad (1, 0, 1), \quad (2, 2, -1).$$

The six sums we need are

$$\sum x_1^2 = 10, \quad \sum x_2^2 = 10, \quad \sum x_3^2 = 4, \quad \sum x_1 x_2 = 9, \quad \sum x_1 x_3 = 0, \quad \sum x_2 x_3 = 0.$$

Check the two zeros by hand, because everything below depends on them: $\sum x_1 x_3 = (-2)(0) + (-1)(-1) + (0)(1) + (1)(1) + (2)(-1) = 0 + 1 + 0 + 1 - 2 = 0$. Dividing each sum by $n - 1 = 4$:

$$S = \begin{pmatrix} 2.5 & 2.25 & 0 \\ 2.25 & 2.5 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (3)$$

Step 1: the characteristic equation. The eigenvalues of S are the λ for which $S - \lambda I$ is singular, that is $\det(S - \lambda I) = 0$:

$$\det \begin{pmatrix} 2.5 - \lambda & 2.25 & 0 \\ 2.25 & 2.5 - \lambda & 0 \\ 0 & 0 & 1 - \lambda \end{pmatrix} = 0.$$

Expand along the third row. Its only non-zero entry is $1 - \lambda$ in position $(3, 3)$, whose cofactor is the top-left 2×2 minor, so the determinant factorises immediately:

$$(1 - \lambda) \left[(2.5 - \lambda)^2 - (2.25)^2 \right] = 0. \quad (4)$$

That factorisation is the block structure showing up in the algebra. A general 3×3 would leave an irreducible cubic; here the zeros in the third row and column hand us one root for free.

Step 2: the three roots. The first factor gives $\lambda = 1$ directly. For the second, expand the bracket:

$$(2.5 - \lambda)^2 - 5.0625 = \lambda^2 - 5\lambda + 6.25 - 5.0625 = \lambda^2 - 5\lambda + 1.1875,$$

and apply the quadratic formula:

$$\lambda = \frac{5 \pm \sqrt{25 - 4(1.1875)}}{2} = \frac{5 \pm \sqrt{20.25}}{2} = \frac{5 \pm 4.5}{2} \implies \lambda = 4.75 \text{ or } 0.25.$$

The discriminant is a perfect square, $\sqrt{20.25} = 4.5$ exactly, which is why this example works with a pen. Multiplying (4) out gives the cubic in full:

$$\lambda^3 - 6\lambda^2 + 6.1875\lambda - 1.1875 = 0. \quad (5)$$

Step 3: check the roots against the invariants. Do not skip this — it catches arithmetic slips before they propagate. For any 3×3 matrix the coefficients of (5) are the three invariants:

$$\begin{aligned}\sum \lambda_j = \text{tr}(S) : & \quad 4.75 + 1 + 0.25 = 6.0, & \text{tr}(S) = 2.5 + 2.5 + 1 = 6.0 \checkmark \\ \sum_{i < j} \lambda_i \lambda_j : & \quad 4.75(1) + 4.75(0.25) + 1(0.25) = 6.1875, & \text{coefficient } 6.1875 \checkmark \\ \prod \lambda_j = \det(S) : & \quad 4.75 \times 1 \times 0.25 = 1.1875, & \det(S) = 1.1875 \checkmark\end{aligned}$$

Step 4: an eigenvector for each root, by solving $(S - \lambda I)v = 0$.

For $\lambda_1 = 4.75$:

$$S - 4.75I = \begin{pmatrix} -2.25 & 2.25 & 0 \\ 2.25 & -2.25 & 0 \\ 0 & 0 & -3.75 \end{pmatrix}.$$

Row 3 reads $-3.75 v_3 = 0$, so $v_3 = 0$. Row 1 reads $-2.25 v_1 + 2.25 v_2 = 0$, so $v_1 = v_2$. Row 2 is row 1 times -1 and adds nothing — the matrix has rank 2, so the solution space is a single line, as it must be for a simple root. Taking $v_1 = 1$ gives $v = (1, 1, 0)$, and dividing by $\|v\| = \sqrt{2}$: $u_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$.

For $\lambda_2 = 1$:

$$S - I = \begin{pmatrix} 1.5 & 2.25 & 0 \\ 2.25 & 1.5 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The top-left 2×2 has determinant $1.5^2 - 2.25^2 = -2.8125 \neq 0$, so rows 1 and 2 force $v_1 = v_2 = 0$. Row 3 is entirely zero, leaving v_3 free. Hence $u_2 = (0, 0, 1)$, already of unit length.

For $\lambda_3 = 0.25$:

$$S - 0.25I = \begin{pmatrix} 2.25 & 2.25 & 0 \\ 2.25 & 2.25 & 0 \\ 0 & 0 & 0.75 \end{pmatrix}.$$

Row 3 gives $v_3 = 0$; row 1 gives $v_1 + v_2 = 0$, so $v_2 = -v_1$. Taking $v_1 = 1$: $u_3 = \frac{1}{\sqrt{2}}(1, -1, 0)$.

Step 5: verify. Multiply each one back through S :

$$S \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2.5 + 2.25 \\ 2.25 + 2.5 \\ 0 \end{pmatrix} = \begin{pmatrix} 4.75 \\ 4.75 \\ 0 \end{pmatrix} = 4.75 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad S \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.25 \\ -0.25 \\ 0 \end{pmatrix} = 0.25 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix},$$

and $S(0, 0, 1)^\top = (0, 0, 1)^\top$. The three are mutually orthogonal — $u_1 \cdot u_3 = \frac{1}{2}(1 - 1 + 0) = 0$, and u_2 has a zero in both of the positions where u_1 and u_3 are non-zero — which is guaranteed, because a symmetric matrix always has an orthogonal eigenbasis.

Sorting by eigenvalue:

$$\begin{aligned}\lambda_1 = 4.75, \quad u_1 &= \frac{1}{\sqrt{2}}(1, 1, 0), & 4.75/6 &= 0.7917 \\ \lambda_2 = 1.00, \quad u_2 &= (0, 0, 1), & 1.00/6 &= 0.1667 \\ \lambda_3 = 0.25, \quad u_3 &= \frac{1}{\sqrt{2}}(1, -1, 0), & 0.25/6 &= 0.0417\end{aligned}$$

Step 6: the scores. These are the centred rows dotted with each u_j . On PC1, u_1 makes the score $(x_1 + x_2)/\sqrt{2}$: $-2.8284, -1.4142, 0.7071, 0.7071, 2.8284$. Their sample variance is $(8 + 2 + 0.5 + 0.5 + 8)/4 = 4.75 = \lambda_1$, as promised. On PC2 the score is just the centred third feature, $(0, -1, 1, 1, -1)$, with variance $4/4 = 1 = \lambda_2$.

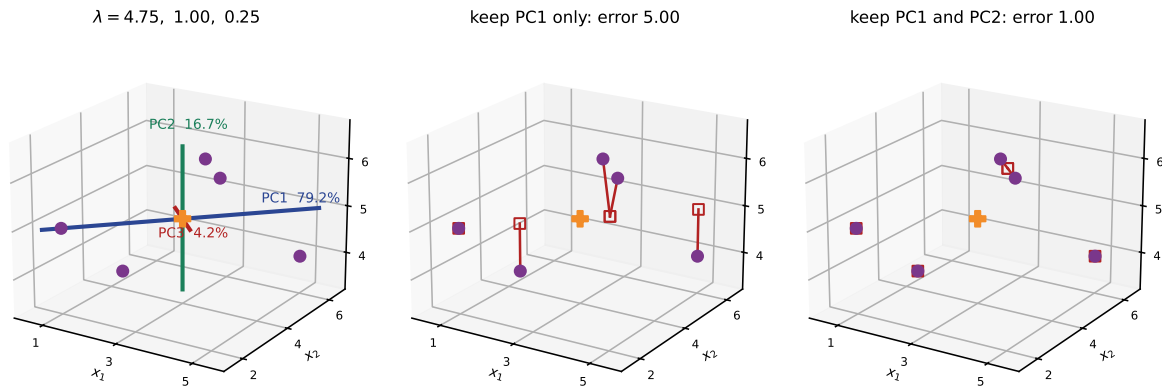


Figure 6: The five points in three dimensions; the vertical axis is x_3 and the orange marker is the mean. Left: the three eigenvectors drawn at length $\sqrt{\lambda_j}$ — PC2 stands vertically, because x_3 is uncorrelated with the other two features. Middle and right: reconstructions from one and from two components, with red lines showing the error on each point.

The shortcut, and why it is worth knowing you are using one

A symmetric 2×2 with *equal* diagonal entries always has eigenvectors $\frac{1}{\sqrt{2}}(1, \pm 1)$ and eigenvalues $a \pm b$, which reads off 4.75 and 0.25 from 2.5 ± 2.25 in one line. That is a genuine shortcut and you may use it — but only once you can produce Steps 1–4 without it, because the moment the diagonal entries differ, or the matrix is not 2×2 after a block split, the shortcut is silent about the answer and the characteristic equation is not.

An uncorrelated feature is already a component

$u_2 = (0, 0, 1)$ is not a mixture of anything — it *is* the third feature. PCA only has work to do where columns are correlated; where they are not, it leaves them alone. That is the honest answer to “what does PCA actually do”: it rotates within the correlated part of the data and ignores the rest.

```
S = np.cov(P - P.mean(axis=0), rowvar=False) # P is 5 x 3
w, V = np.linalg.eigh(S) # by hand
p = PCA(n_components=3).fit(P) # by library
for k in (1, 2):
    pk = PCA(n_components=k).fit(P)
    R = pk.inverse_transform(pk.transform(P))
```

```
covariance matrix
[[2.5  2.25  0. ]
 [2.25 2.5  0. ]
 [0.   0.   1. ]]
eigenvalues [4.75 1.  0.25]  ratios [0.791667 0.166667 0.041667]
sklearn components_ (rows)
[[ 0.707107  0.707107  0.      ]
 [ 0.         0.         1.      ]
 [ 0.707107 -0.707107 -0.      ]]
k=1 error 5.000000 = (n-1)*sum(lambda_{j>k}) = 5.000000  (4 distinct points of 5)
k=2 error 1.000000 = (n-1)*sum(lambda_{j>k}) = 1.000000  (4 distinct points of 5)
```

Three things to notice. The library’s second component is exactly $(0, 0, 1)$, the same answer the block structure gave us by hand. Both reconstructions collapse five points onto four distinct ones, because $(3, 5, 6)$ and $(4, 4, 6)$ differ only along u_3 : dropping a component **merges points that differed only along it**, which is what “lossy” means. And both errors match $(n - 1) \sum_{j>k} \lambda_j$ exactly — $4(1.00 + 0.25) = 5.00$ and $4(0.25) = 1.00$ — which is the general identity

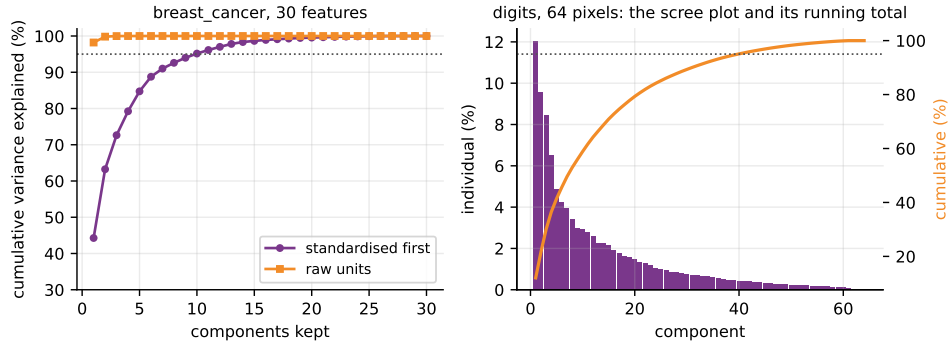


Figure 7: Left: cumulative variance explained for breast_cancer, with and without standardising first. Right: the digits scree plot with its running total on the right-hand axis.

of the next section.

6.3 The reconstruction identity, and choosing k

Write the centred data in the eigenbasis: $x_c = \sum_j t_j u_j$ with $t_j = x_c^\top u_j$. Keeping the top k components means setting $t_{k+1}, \dots, t_d$ to zero, so the squared error for that point is $\sum_{j>k} t_j^2$. Summing over all n points and using $\sum_i t_{ij}^2 = (n-1)\lambda_j$:

$$\text{total squared reconstruction error} = (n-1) \sum_{j>k} \lambda_j, \quad \frac{\text{error}}{\text{total}} = 1 - \frac{\sum_{j \leq k} \lambda_j}{\sum_j \lambda_j}. \quad (6)$$

So *variance kept* and *relative error removed* add to exactly one. That is why a single curve answers both questions.

```
breast_cancer, standardised, 30 features
  1 components explain 44.27%    80% needs 5 components (16.7% of 30)
  2 components explain 63.24%    90% needs 7 components (23.3% of 30)
  5 components explain 84.73%    95% needs 10 components (33.3% of 30)
 10 components explain 95.16%    99% needs 17 components (56.7% of 30)
digits, standardised, 64 features
80% needs 21  90% needs 31  95% needs 40  99% needs 54  of 64
```

Caution

There is no universal “keep 95%” rule. On breast_cancer that is ten of thirty columns, because the thirty features are ten measurements each recorded three ways and are therefore heavily redundant. On digits it is forty of sixty-four. **95% of what depends entirely on the data.** An elbow in the scree plot is a suggestion, not a criterion; if k matters to your score, choose it by cross-validation like any other hyper-parameter.

The reconstruction is worth seeing rather than reading.

```
digits, PCA reconstruction error against k
k= 1  variance kept 14.89%  relative reconstruction error 85.11%
k= 2  variance kept 28.51%  relative reconstruction error 71.49%
k= 5  variance kept 54.50%  relative reconstruction error 45.50%
k=10  variance kept 73.82%  relative reconstruction error 26.18%
k=20  variance kept 89.43%  relative reconstruction error 10.57%
k=30  variance kept 95.91%  relative reconstruction error  4.09%
k=40  variance kept 98.82%  relative reconstruction error  1.18%
k=64  variance kept 100.00% relative reconstruction error  0.00%
```

Every row sums to 100%, which is Equation (6) checked numerically eight times.

6.4 Scale before you rotate

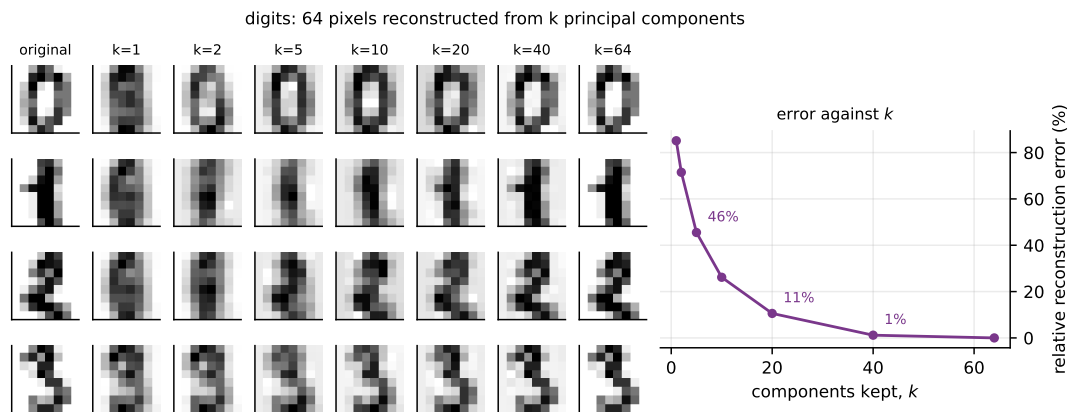


Figure 8: Four digits reconstructed from k principal components. At $k = 5$ you can tell a 0 from a 1; at $k = 20$ they are all legible and 44 of the 64 numbers per image have been discarded.

```
PCA on breast_cancer WITHOUT scaling
PC1 explains 0.9820 of the variance
its largest loading is 0.8521 on 'worst area'
that feature's standard deviation is 568.9; the smallest in the table is 0.00264
PCA on breast_cancer WITH StandardScaler
PC1 explains 0.4427 of the variance
its largest loading is 0.2609 on 'mean concave points'
```

Unscaled, the first component explains 98.20% of the variance and is essentially the **worst area** column with a new name. That is not a discovery; it is the definition of PCA meeting a table whose columns have different units. Variance has units, and the widest column wins by construction.

Key idea

Standardise before PCA unless every column is already measured in the same unit. Otherwise you have run an eigendecomposition to find out which of your columns you recorded in the biggest numbers.

6.5 PCA is not feature selection

```
PC1 of the standardised breast_cancer data: 30 loadings
largest |loading| 0.2609   smallest |loading| 0.0145
number of features with |loading| > 0.10 : 26 of 30
number needed to reach 90% of the loading vector's length: 20
+0.2609 mean concave points +0.2584 mean concavity
+0.2509 worst concave points +0.2393 mean compactness
to compute PC1 for a new patient you need ALL 30 measurements.
```

Caution

“I reduced to ten components” does not mean “I only need ten measurements”. Every component is a weighted sum of all thirty original columns, so the cost of collecting one row is unchanged. **If your reason for reducing was that a test is expensive or invasive, PCA has not helped you at all;** feature selection has. And $0.2609 \times \text{concave points} + 0.2584 \times \text{concavity} + \dots$ is not a quantity a clinician can name, so interpretability is gone too.

6.6 PCA has never heard of your labels

```
rng = np.random.default_rng(0) # the one seed for this lecture
big = rng.normal(0, 10.0, 600) # huge variance, no signal
```

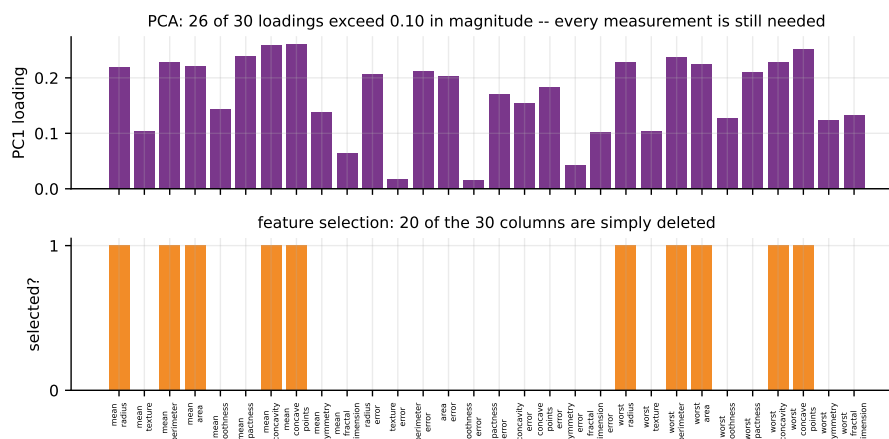


Figure 9: Top: the thirty PC1 loadings, twenty-six of them above 0.10 in magnitude. Bottom: what a filter selecting ten features does to the same thirty columns. One rotates; the other deletes.

```
small = rng.normal(0, 1.0, 600) # small variance, IS the label
y = (small > 0).astype(int)
X = np.c_[big, small]
```

```
PC1 explains 0.9909 of the variance; its loadings are [1.      0.0052]
logistic regression on both features      accuracy 0.9850
logistic regression on PCA's 1 component  accuracy 0.5200
logistic regression on the 1 feature a filter would pick 0.9933
after standardising first, PCA's component accuracy 0.7617
```

PCA discarded the only informative direction and kept the noise, taking the accuracy from 0.9850 to a coin flip. Standardising first helps a little (0.7617) but does not fix it, because the problem is not the units — it is that y was never consulted. A supervised filter picking one column scored 0.9933.

Key idea

PCA is **unsupervised**: it maximises variance, and variance is not relevance. In practice it usually works, because on real data the directions that vary most usually do carry the signal — but “usually” is not “always”, and you should check rather than assume.

Two named alternatives worth knowing. **Linear discriminant analysis** finds directions that separate the *classes* rather than maximising variance, and would solve the example above. **Partial least squares** (ISLP §6.3) does the analogous thing for regression. Both are supervised, and both therefore must be fitted inside the fold.

6.7 PCA is fitted on the training split

```
components whose sign flipped between the two fits: 3.0 of 10, up to 6
correct          accuracy 0.9768 +/- 0.0128  worst split 0.9357
fit on everything  accuracy 0.9773 +/- 0.0122  worst split 0.9415
refit on the test set  accuracy 0.9447 +/- 0.0226  worst split 0.8947
refitting on the test set costs -0.0320 on average and -0.0877 on the
worst of the 40 splits
fit-on-everything minus correct: +0.0006  t +0.94  wins 8/40
```

Be honest about which of the two mistakes is which. Fitting the scaler and PCA on all the rows before splitting leaked **nothing measurable** here: +0.0006 with $t = 0.94$ over 40 splits, and the leaky version won on only 8 of them. That is what we should expect — PCA never sees y , so the only thing it can absorb from the test rows is a little information about the covariance structure.

Calling `fit_transform` *again* on the test set is a different and much worse error, and it is the one students actually make. The test components are a different basis: on average three of the ten point the opposite way, so “column 3” in training and “column 3” at test time are not the same feature. That cost -0.0320 on average, and -0.0877 on the worst of the forty splits.

Caution

Use `fit_transform` on the training data and `transform` — never `fit_transform` — on anything else. A `Pipeline` enforces this for you, which is the entire reason it exists.

The reason PCA leaks so little is the one Section 6.6 gave: it is unsupervised. The step that leaks a great deal is the one that *does* look at y , and it is next.

Sources: Han, Kamber and Pei, *Data Mining: Concepts and Techniques* (3e), Ch. 3.; Müller and Guido, *Introduction to Machine Learning with Python*, Ch. 3–4.; James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*, Ch. 5–6, 12.; Scikit-learn documentation (preprocessing, model selection), accessed 2026-08-04.

7 Feature selection

7.1 Three families

	How it decides	Cost and risk
Filter	a statistic per feature, computed once, model ignored: variance, ANOVA F , mutual information, correlation, χ^2	cheapest; blind to interactions and to redundancy between features
Wrapper	fit the model repeatedly and keep what helps: forward selection, backward elimination, recursive feature elimination	most expensive; can overfit the selection itself
Embedded	the model selects while it fits: L1 (lasso), tree importances	nearly free; the answer is specific to that model

Han, Kamber and Pei call this **attribute subset selection**, and list it under dimensionality reduction alongside PCA. The difference from PCA is the one Section 6.5 made: selection deletes columns, so the row you have to collect gets cheaper and the model stays readable.

7.2 Filter, part 1: variance

```
VarianceThreshold(threshold=0.0).fit(Xg)
```

```
digits: 3 of the 64 pixels have zero variance (always the same value)
       48 of the 64 survive a variance threshold of 1.0
logistic regression on all 64 pixels          accuracy 0.9694
threshold 0.0 keeps 61 pixels                accuracy 0.9694
threshold 1.0 keeps 48 pixels                accuracy 0.9694
threshold 10.0 keeps 43 pixels               accuracy 0.9683
```

Sixteen of the sixty-four columns removed for nothing at all, and twenty-one for 0.0011. A variance filter almost never improves a score — what it buys is **columns**. It is also a correctness measure rather than a performance one: a constant column makes `StandardScaler` divide by zero and makes every correlation with it undefined.

Note the unit problem: variance has units, so a threshold of 1.0 means something different for metres and for millimetres. Threshold each column relative to its own scale, or use a scale-free filter.

7.3 Filter, part 2: the F test and mutual information disagree

The ANOVA F statistic for classification compares the *class means* of a feature. Mutual information estimates $I(X_j; Y)$ and makes no assumption about the form of the dependence. Where the dependence is not a difference of means, they part company.

```
rng = np.random.default_rng(0) # the one seed for this lecture
a = rng.normal(0, 1, 1000) # enters the label linearly
b = rng.normal(0, 1, 1000) # enters only through |b|
noise = rng.normal(0, 1, (1000, 8))
lab = ((a + (np.abs(b) - 0.8) * 2.0 + rng.normal(0, 0.4, 1000)) > 0).astype(int)
```

feature	ANOVA F	rank	mutual info	rank
a (linear)	317.110	1	0.1342	2
b (b only)	0.213	8	0.2263	1
best noise	8.326	-	0.0123	-
b ranks 8 of 10 by F and 1 of 10 by mutual information.				
SelectKBest(f_classif, k=3)			picks ['a', 'noise 1', 'noise 2']	
SelectKBest(mutual_info_classif, k=3)			picks ['a', 'b', 'noise 8']	

The two classes have the same mean b — the label depends on $|b|$, which is symmetric — so the F test sees an F of 0.213 and ranks the genuinely informative feature *eighth of ten*, below six of the eight noise columns. Mutual information ranks it first.

Key idea

A filter measures the kind of dependence it was built to measure, and nothing else. Choosing a filter is choosing an assumption about how features relate to the target.

There is a price, and it is worth stating in a form that does not depend on the machine. Mutual information is non-parametric: for every one of the 30 columns, in every one of the 5 folds, it runs a nearest-neighbour search — 150 searches — where the F test needs two sums per column. It also needs more rows to estimate reliably. On the machine these notes were built on that came to 0.23 seconds against 0.02; on yours it will be different numbers with the same ordering.

7.4 Wrapper: recursive feature elimination

Fit the model. Drop the feature with the smallest coefficient. Refit on what remains. Repeat. With `step=1` on 30 features that is 30 model fits.

```
RFE(LogisticRegression(max_iter=5000), n_features_to_select=10).fit(Z, y)
```

```
30 features with step=1 means 30 fits.
elimination order, first six      the last one standing: worst area
dropped 30 of 30: worst compactness    dropped 27: mean smoothness
dropped 29 of 30: mean symmetry        dropped 26: texture error
dropped 28 of 30: concavity error      dropped 25: concave points error
RFE k= 1 accuracy 0.9157      RFE k=10 accuracy 0.9684
RFE k= 2 accuracy 0.9526      RFE k=20 accuracy 0.9789
RFE k= 5 accuracy 0.9684      RFE k=30 accuracy 0.9789
RFECV picks its own k: 26 features, accuracy 0.9824
```

A wrapper asks the *model* what it needs, so it can see redundancy that a univariate filter cannot: if two columns carry the same information, a filter scores both highly and keeps both, while

RFE drops one as soon as the other is present. The price is a model fit per step, and the answer is that model's answer — RFE with a logistic regression and RFE with a random forest will disagree.

RFECV runs RFE inside a cross-validation loop and picks the k that maximises the score. Left to itself it kept 26 of the 30, which is the first hint of Section 7.7.

7.5 Embedded: L1 shrinks coefficients to exactly zero

Lecture 12 treats regularisation properly; this section needs only enough of it to use the lasso as a feature selector.

Coefficients and what a zero means. A linear model predicts $\hat{y} = w_0 + w_1x_1 + \dots + w_dx_d$. Each w_j is a **coefficient**: how much the prediction moves when feature j rises by one unit. If $w_j = 0$ the feature is multiplied by zero and plays no part in any prediction. A model with some coefficients at exactly zero has therefore *selected* the others, and counting the non-zero coefficients counts the features kept.

Penalties. Plain least squares minimises only the prediction error. A **penalised** model adds a charge for large coefficients, with a strength $\alpha \geq 0$ that you choose:

$$\text{ridge (L2 penalty): } \min_w \sum_i (y_i - \hat{y}_i)^2 + \alpha \sum_j w_j^2, \quad (7)$$

$$\text{lasso (L1 penalty): } \min_w \sum_i (y_i - \hat{y}_i)^2 + \alpha \sum_j |w_j|. \quad (8)$$

The names come from the two ways of measuring the size of w : the L2 norm uses squares, the L1 norm absolute values. At $\alpha = 0$ both are ordinary least squares; as α grows, the coefficients are pulled towards zero. Refitting the same model at a range of α values traces out a **path**, which is what the output below does.

Why only L1 produces exact zeros. Near zero, the square w_j^2 becomes tiny, so the ridge penalty stops pushing a coefficient that is already small: ridge makes every coefficient smaller but almost never exactly zero. The absolute value $|w_j|$ charges the same α per unit of w_j all the way down to zero, so a feature whose contribution to the fit is worth less than that charge is set to exactly 0. That is why the lasso *selects* and ridge only *shrinks*. Because the selection happens as part of fitting, rather than in a separate step before (a filter) or around (a wrapper) the model, it is called **embedded** selection.

Why standardise first. The penalty adds up $|w_j|$ across features. A feature measured in large units needs only a small coefficient, so it is penalised less for the same real effect. Standardising puts every feature on the same scale, which makes the comparison fair (Section 3 made the same point for distances and gradients).

The data. `load_diabetes` has 442 patients and ten baseline measurements: `age`, `sex`, `bmi` (body mass index), `bp` (average blood pressure) and six blood-serum readings — `s1` total cholesterol, `s2` LDL, `s3` HDL, `s4` total cholesterol over HDL, `s5` the log of serum triglycerides and `s6` blood sugar. The target is a quantitative measure of disease progression one year later. In the output, `nonzero` is the number of coefficients the lasso kept (named when there are six or fewer) and `CV R2` is the mean R^2 (Equation (2)) over five-fold cross-validation, with the scaler and the lasso refitted inside each fold.

scikit-learn scales the lasso's error term

Lasso actually minimises $\frac{1}{2n} \sum_i (y_i - \hat{y}_i)^2 + \alpha \sum_j |w_j|$, while **Ridge** uses Equation (7) as written. The same number α therefore means a much stronger penalty in **Lasso** than in **Ridge**; compare each model with itself across α , not the two at one value.

```
for alpha in (0.01, 0.1, 0.5, 1.0, 2.0, 5.0, 20.0):
    la = Lasso(alpha=alpha).fit(StandardScaler().fit_transform(Xd), yd)
    print(alpha, (la.coef_ != 0).sum())
```

```
alpha 0.01 10 nonzero CV R2 +0.4823
alpha 0.10 9 nonzero CV R2 +0.4825
alpha 0.50 8 nonzero CV R2 +0.4818
alpha 1.00 7 nonzero CV R2 +0.4820
alpha 2.00 7 nonzero CV R2 +0.4798
alpha 5.00 5 nonzero CV R2 +0.4659 sex,bmi,bp,s3,s5
alpha 20.00 3 nonzero CV R2 +0.3492 bmi,bp,s5
ridge alpha 1.00 10 nonzero CV R2 +0.4822
```

Five features of ten retain 97% of the R^2 ; three retain 72%. The lasso hands you the whole trade-off curve for free and α is the dial. Ridge ($\alpha = 1$) keeps all ten and scores the same: **L1 selects, L2 only shrinks**. Because the selection depends on α , the selection is a hyper-parameter and belongs inside the cross-validation loop like any other.

7.6 The four selectors do not agree

```
ANOVA F      picked 10: [0, 2, 3, 6, 7, 20, 22, 23, 26, 27]
mutual info   picked 10: [0, 2, 3, 6, 7, 13, 20, 22, 23, 27]
RFE           picked 10: [7, 10, 13, 15, 20, 21, 22, 23, 26, 27]
L1 logistic   picked 8: [7, 10, 20, 21, 24, 26, 27, 28]
```

pairwise overlap (Jaccard)

```
ANOVA F      vs mutual info   9 shared, Jaccard 0.82
ANOVA F      vs RFE           6 shared, Jaccard 0.43
ANOVA F      vs L1 logistic    4 shared, Jaccard 0.29
mutual info   vs RFE           6 shared, Jaccard 0.43
mutual info   vs L1 logistic    3 shared, Jaccard 0.20
RFE           vs L1 logistic    6 shared, Jaccard 0.50
```

```
chosen by all four: mean concave points, worst radius, worst concave points
chosen by at least one: 16 of 30 features
```

Three features of thirty command a consensus. The two filters agree with each other (0.82) and with nobody else, which makes sense — they are both univariate. Sixteen of the thirty columns were picked by somebody.

Key idea

There is no such thing as “the important features”. There is only “the features that this criterion, with this model, at this k , chose”. Report the criterion whenever you report a selection.

7.7 Does selection help? Report the experiment honestly

```
selector      accuracy
all 30 features 0.9789
filter: ANOVA F, k=10 0.9596
filter: mutual info, k=10 0.9561
wrapper: RFE, k=10 0.9684
embedded: L1, C=0.08 0.9719
cost in counts, which are the same on every machine:
```

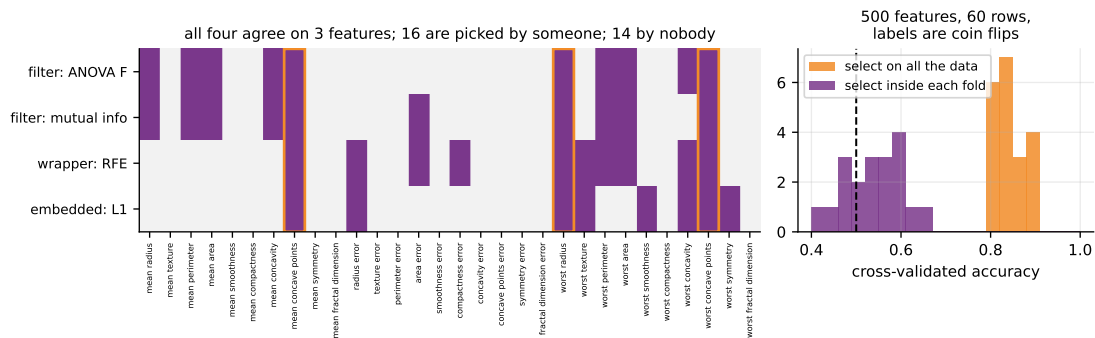


Figure 10: Left: the four selectors against the thirty breast_cancer features; orange boxes mark the three on which all four agree. Right: the same experiment on the same random data, differing only in whether the selector was fitted inside the fold.

RFE k=10	21 model fits per fold (30 features down to 10, one at a time) = 105 fits over 5 folds
ANOVA F k=10	1 model fit per fold = 5 fits over 5 folds, so RFE costs 21 times as many fits

Not one of the four beat simply using all thirty columns. RFE, at 21 model fits per fold against the filter's one, recovered the most of the loss. This is not a failure of the experiment; it is the answer.

Count fits, not seconds

Everything else in these notes reproduces exactly. The seconds do not: they depend on your processor, your BLAS build and what else is running, and the same comparison on the same machine varies from run to run. So the cost claim above is stated in *model fits* — 21 per fold against 1, true on every machine — and the seconds are reported only as a labelled measurement. If you time it and get a different ratio, you are not wrong.

Caution

Do not select features because you were told that selecting features is good practice. Select when you need a cheaper row to collect, a model somebody has to read, a model small enough to deploy, or when d is large relative to n . At 569×30 , breast_cancer is none of those things.

7.8 Selecting with the labels, before the split

This is the centrepiece of the lecture.

```
for rep in range(20): # a sweep: 20 different draws
    rng = np.random.default_rng(100 + rep)
    Xr = rng.normal(size=(60, 500)) # pure noise
    yr = rng.integers(0, 2, 60) # labels are coin flips

    # wrong: score every feature on ALL the data, then cross-validate
    f, _ = f_classif(Xr, yr)
    keep = np.argsort(-f)[:20]
    cross_val_score(LogisticRegression(), Xr[:, keep], yr, cv=5).mean()

    # right: the selector is a step in the Pipeline
    cross_val_score(Pipeline([("sel", SelectKBest(f_classif, k=20)),
                              ("lr", LogisticRegression())]),
                    Xr, yr, cv=5).mean()
```

```

500 features, 60 rows, labels are pure coin flips. Truth is 0.5000.
  select on ALL the data, then cross-validate : 0.8475 +/- 0.0293
  select INSIDE each fold (Pipeline)          : 0.5342 +/- 0.0710
  overstatement                               +0.3133
  the wrong way beat 0.5 on 20 of 20 repetitions

```

There is nothing to learn in this dataset. The honest answer is 0.5. The first procedure reports 0.85, every single time.

The mechanism is worth stating slowly. With 500 candidates and 60 rows, some features correlate with the labels by chance — that is what 500 tries at a coin-flipping game buys you — and the F test finds them. But it found them *using the labels of every row, including the rows that will shortly be held out*. By the time cross-validation runs, the held-out rows have already voted on which columns are in the model.

Key idea

Any step that looks at y is part of the model and must be refitted inside every fold. Scaling and PCA leak little because they never see y . Feature selection sees y , and it leaked +0.3133.

This is the classic error described in Hastie, Tibshirani and Friedman's *Elements of Statistical Learning* §7.10.2 as “the wrong way to do cross-validation”; it is the reason a great many published results in high-dimensional biology did not replicate.

7.9 Everything in one pipeline

```

pipe = Pipeline([
    ("scale", StandardScaler()),
    ("select", SelectKBest(f_classif, k=15)),
    ("pca", PCA(n_components=8)),
    ("model", LogisticRegression(max_iter=5000)),
])
cross_val_score(pipe, Xb, yb, cv=5).mean()

```

```

scale -> select 15 -> PCA to 8 -> logistic regression
5-fold accuracy 0.9490 +/- 0.0140
compare: raw logistic regression, all 30 features 0.9789

```

The mean and standard deviation of each column, the fifteen chosen columns and the eight rotation directions are all recomputed inside every fold from four fifths of the data. Nothing leaks.

And read the second line, because it is the honest end of the lecture: the four-step pipeline scored *worse* than doing nothing at all. It cannot leak, and it also did not help. Both facts are worth reporting.

Sources: Han, Kamber and Pei, *Data Mining: Concepts and Techniques* (3e), Ch. 3.; Müller and Guido, *Introduction to Machine Learning with Python*, Ch. 3–4.; James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*, Ch. 5–6, 12.; Scikit-learn documentation (preprocessing, model selection), accessed 2026-08-04.

8 Summary

1. A representation is good only relative to a model. One engineered product column took a linear model from $R^2 = 0.9091$ to 0.9981; a random forest reached the same place unaided. Binning bought a linear model +0.0957 and cost a tree −0.0323, and afterwards both fitted exactly the same step function.

2. `PolynomialFeatures` degree 3 turns 30 columns into 5455. On diabetes, degree 2 cost -0.0643 of R^2 even with ridge.
3. Age against income: the income column supplied 99.995% of the squared distance, and standardising **changed which customer was the nearest neighbour**.
4. Scaling was worth $+0.0561$ to an RBF SVM, $+0.0334$ to k -NN, $+0.0246$ to logistic regression, and **exactly** 0.0000 to a decision tree and a random forest — the same tree, node for node, because a split is a question about order.
5. Unscaled, stochastic gradient descent diverged at every usable step size ($R^2 = -1.7 \times 10^{28}$) and stalled at 0.35 at the small ones. Standardised, it reached 0.5144 against a least-squares 0.5177 in fourteen epochs. Condition number: 1.03×10^6 against 470.
6. One typo squeezed the good data $78\times$ under `StandardScaler`, $71\times$ under `MinMaxScaler` and $2\times$ under `RobustScaler`.
7. Ordinal-coding a nominal variable gave $R^2 = 0.0743$ against 0.9868 for one-hot; the hand calculation predicted 0.0577, and a tree on the same ordinal column got 1.0000. An intercept plus all k dummies is rank-deficient (condition number 3.4×10^{15}), and two completely different coefficient vectors gave identical predictions.
8. One-hot encoding a 256-level identifier that carried no information moved the training R^2 up 0.14 and the test R^2 down 0.12.
9. In 1000 dimensions the farthest of 499 points is only $1.11\times$ as far as the nearest. Sampling half the digits rows cost 0.0087 of accuracy; sampling a twentieth cost 0.1188.
10. PCA on unscaled breast_cancer: PC1 explained 98.20% and was the widest column. Standardised: 44.27%, and 95% needed 10 of 30 components — but 40 of 64 on digits. Variance kept and relative reconstruction error always sum to 100%.
11. PCA is a rotation, not a selection: 26 of 30 loadings exceed 0.10, so all thirty measurements are still needed. And it is unsupervised — it took an accuracy of 0.9850 down to 0.5200 by discarding the only informative direction.
12. Refitting PCA on the test set cost -0.0320 on average and -0.0877 at worst; fitting it on all the rows before splitting cost nothing measurable ($t = 0.94$). **Selecting features with the labels before cross-validating scored 0.8475 on pure coin flips, against 0.5342 inside a Pipeline.**
13. None of the four selectors beat using all thirty features on breast_cancer, and RFECV chose to keep 26 of them. Report that kind of result rather than hiding it.

9 Practice questions

Attempt these before reading Section 10. Questions 1, 2, 3, 5 and 6 are pure hand arithmetic and are the ones most likely to appear on a paper.

1. A column contains 10, 20, 30, 40, 100. (a) Compute its mean, population standard deviation, minimum, maximum, median, Q_1 , Q_3 and IQR. (b) Write out the column after `StandardScaler`, `MinMaxScaler` and `RobustScaler`. (c) What fraction of the $[0, 1]$ interval do the first four values occupy after min-max scaling, and what does that imply for a k -NN model?

2. Three people are recorded as height in cm, weight in kg and hours of exercise per week: $P = (170, 60, 3)$, $Q = (155, 62, 9)$, $R = (172, 85, 4)$. (a) Compute $d(P, Q)^2$ and $d(P, R)^2$ on the raw numbers, showing the contribution of each column. (b) Standardise each column and repeat. (c) Which point is P 's nearest neighbour in each case, and which answer would you defend?
3. A nominal column `region` takes the values `north`, `south`, `east`, `west`, whose mean targets are 40, 40, 10, 10 respectively. (a) Write down the codes `OrdinalEncoder` will assign with its default settings. (b) Fit the best straight line through the four (code, mean) points by hand and give its R^2 . (c) What is the R^2 of a one-hot fit? (d) If you one-hot encode all four levels and include an intercept, what is the rank of the resulting 4×5 design matrix, and why does that matter?
4. You have a column of shirt sizes `small`, `medium`, `large` and you write `OrdinalEncoder().fit_transform(s`. What codes do you get, and why? Give the one-line fix. Why does nothing raise an error?
5. A table has three nominal columns with 50 levels each and 2000 rows. (a) How many columns after one-hot encoding, with and without `drop='first'`? (b) How many degree-2 interaction terms would `PolynomialFeatures` then generate from those one-hot columns alone? (c) Given the measured result of Section 4.4, what would you do instead?
6. A centred two-feature dataset has covariance matrix $S = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$. (a) Find both eigenvalues and eigenvectors by hand. (b) What fraction of the variance does PC1 explain? (c) Check your eigenvalues against the trace. (d) If the dataset has $n = 21$ points, what is the total squared reconstruction error from keeping only PC1?
7. Prove Equation (6): that the total squared reconstruction error from keeping the top k components is $(n - 1) \sum_{j>k} \lambda_j$. Then use the digits table in Section 6.3 to state, without any further computation, the variance explained by the top 20 of the 64 components.
8. You must reduce a 40-column clinical panel because two of the tests cost a great deal to run. A colleague proposes PCA to 10 components. Explain in three sentences why this does not solve the problem, cite one measured number from Section 6.5, and say what you would do instead.
9. A colleague reports 0.85 cross-validated accuracy on a gene-expression table with 500 columns and 60 patients. Their notebook, in order: (i) `load`; (ii) `MinMaxScaler().fit_transform(X)`; (iii) compute the ANOVA F of every column against `y` and keep the best 20; (iv) `cross_val_score(LogisticRegression(X_kept, y, cv=5))`. Identify every problem, say which one accounts for essentially all of the number, quantify it from this lecture, and give the corrected code.
10. You are handed a table of 50,000 customer records: six numeric columns whose scales differ by four orders of magnitude, one ordinal column (`bronze/silver/gold`), one nominal column with 12 levels, one city column with 1800 levels, and a binary target. You must ship a model that a business analyst will read. Design the preprocessing. For each decision cite a measured number from this lecture.

10 Solutions

1. (a) Mean = $200/5 = 40$. Deviations $-30, -20, -10, 0, 60$; their squares sum to $900 + 400 + 100 + 0 + 3600 = 5000$; dividing by $n = 5$ gives 1000, so $\sigma = 31.6228$. Median = 30, $Q_1 = 20$, $Q_3 = 40$, IQR = 20.

(b) $z: (10-40)/31.6228 = -0.9487$ and so on. Min-max: $(10-10)/(100-10) = 0$, $(20-10)/90 = 0.1111$, and so on. Robust: $(10-30)/20 = -1$, $(20-30)/20 = -0.5$, and so on.

```
x      [ 10.  20.  30.  40. 100.]
mean 40.0000 sd(ddof=0) 31.6228
min 10.0 max 100.0 median 30.0 Q1 20.0 Q3 40.0 IQR 20.0
StandardScaler [-0.9487 -0.6325 -0.3162  0.      1.8974]
MinMaxScaler  [0.      0.1111 0.2222 0.3333 1.      ]
RobustScaler  [-1.    -0.5   0.     0.5   3.5]
MinMax: the first four values occupy [0.0000, 0.3333] -- 33.3% of the [0,1] range
```

(c) The four ordinary values occupy a third of the range; with a genuine outlier (Section 3.4 used 500 against values around 5) it becomes 1.3%. For k -NN that means every ordinary point becomes a near-duplicate of every other along that column, while the outlier sits in its own universe. Note that `RobustScaler` is the only one of the three that leaves the four ordinary values evenly spaced.

2. (a) Raw: $d(P,Q)^2 = 15^2 + 2^2 + 6^2 = 225 + 4 + 36 = 265$, of which height is 84.9%; $d(P,R)^2 = 2^2 + 25^2 + 1^2 = 4 + 625 + 1 = 630$, of which weight is 99.2%.

```
raw    d(P,Q)^2 = 225.0 + 4.0 + 36.0 = 265.0
raw    d(P,R)^2 = 4.0 + 625.0 + 1.0 = 630.0
standardised:
  P  +0.5712 -0.7934 -0.8890
  Q  -1.4060 -0.6171 +1.3970
  R  +0.8348 +1.4105 -0.5080
scaled d(P,Q)^2 = 3.9093 + 0.0311 + 5.2258 = 9.1662
scaled d(P,R)^2 = 0.0695 + 4.8575 + 0.1452 = 5.0722
nearest neighbour of P: raw Q, scaled R
```

(c) Raw, P 's nearest neighbour is Q ; standardised, it is R . Defend the standardised answer: on the raw numbers the three columns span 17 cm, 25 kg and 6 hours, so weight was silently given four times the influence of exercise, and switching to grams or minutes would have changed the answer again. Standardising makes the weighting explicit and unit-free.

3. (a) `OrdinalEncoder` sorts alphabetically: `east = 0`, `north = 1`, `south = 2`, `west = 3`. So the (code, mean) pairs are (0, 10), (1, 40), (2, 40), (3, 10).

(b) $\bar{x} = 1.5$, $\bar{y} = 25$. The x -deviations are $-1.5, -0.5, +0.5, +1.5$ and the y -deviations $-15, +15, +15, -15$, so

$$S_{xy} = 22.5 - 7.5 + 7.5 - 22.5 = 0.$$

The slope is **exactly zero**: the best straight line is the flat line $y = 25$, and $R^2 = 0$.

```
alphabetical codes: {'north': 1, 'south': 2, 'east': 0, 'west': 3}
fitted line y = 25.0000 + 0.0000 * code
predictions [25. 25. 25. 25.] against truth [40. 40. 10. 10.]
R2 of the ordinal fit 0.0000
R2 of the one-hot fit 1.0000
intercept + 4 dummies: shape (4, 5) rank 4 (needs 5)
with drop='first': rank 4 of 4 columns
```

(c) One-hot gives $R^2 = 1$: with one indicator per level the model has one free parameter per group mean.

(d) The 4×5 matrix has rank 4, not 5, because the four dummy columns sum to the intercept column. The coefficients are therefore not unique — Section 4.3 exhibits two vectors giving identical predictions — so no coefficient, standard error or p -value from that fit means anything. Use `drop='first'`.

4. You get `large = 0`, `medium = 1`, `small = 2`, because the default is to sort the observed categories as strings and `"large" < "medium" < "small"` alphabetically. Your genuinely ordinal variable has been given the reverse of its real order, with `large` and `small` at the two ends and the sign of every coefficient flipped.

Running it on `small`, `medium`, `large`, `medium`, `small` prints `[2. 1. 0. 1. 2.]` with `categories_ = ['large' 'medium' 'small']`; with the order stated it prints `[0. 1. 2. 1. 0.]`.

The fix is one argument: `OrdinalEncoder(categories=[["small","medium","large"]])`. Nothing raises because nothing is wrong from the encoder's point of view — it was asked to turn strings into integers and it did. This is the most common silent bug in the whole of preprocessing: always print `enc.categories_` after fitting.

5. (a) $3 \times 50 = 150$ columns, or $3 \times 49 = 147$ with `drop='first'`.

(b) $\binom{150}{2} = 150 \times 149/2 = 11,175$ pairwise products, of which the great majority are identically zero (two levels of the same column can never both be 1).

(c) 150 columns on 2000 rows is survivable but wasteful, and it is exactly the regime where our measured experiment lost 0.11 of test R^2 to a meaningless identifier. Group levels below a frequency cut-off into `other`; or use a count or target encoding fitted inside the fold; or use a gradient-boosted tree implementation with native categorical support and skip the encoding entirely. Do not generate the 11,175 interactions.

6. (a) S is symmetric with equal diagonal entries, so as in Section 6.2 the eigenvectors are $\frac{1}{\sqrt{2}}(1, 1)$ and $\frac{1}{\sqrt{2}}(1, -1)$ with eigenvalues $4 + 2 = 6$ and $4 - 2 = 2$. If you prefer the characteristic polynomial: $(4 - \lambda)^2 - 4 = 0 \Rightarrow \lambda = 4 \pm 2$.

(b) $6/(6 + 2) = 0.75$.

(c) $\text{tr}(S) = 4 + 4 = 8 = 6 + 2$.

(d) By Equation (6), $(n - 1)\lambda_2 = 20 \times 2 = 40$.

```
eigenvalues [6. 2.]    eigenvectors [[0.7071, -0.7071], [0.7071, 0.7071]]
PC1 explains 0.7500    trace 8.0 = sum of eigenvalues 8.0
```

7. Write each centred point in the eigenbasis, $x_c = \sum_{j=1}^d t_j u_j$ with $t_j = x_c^\top u_j$; this is legitimate because the eigenvectors of a symmetric matrix form an orthonormal basis. Keeping the top k replaces x_c by $\hat{x}_c = \sum_{j \leq k} t_j u_j$, so the residual is $\sum_{j > k} t_j u_j$ and, by orthonormality,

$$\|x_c - \hat{x}_c\|^2 = \sum_{j > k} t_j^2.$$

Sum over the n points. The scores on component j have mean zero and sample variance λ_j , so $\sum_i t_{ij}^2 = (n - 1)\lambda_j$, giving $(n - 1)\sum_{j > k} \lambda_j$. Dividing by the same expression with $k = 0$ gives the relative form, and since the two fractions sum to one, **variance kept + relative error = 100%**.

From the digits table, $k = 20$ has a relative reconstruction error of 10.57%, so the top 20 components explain $100 - 10.57 = 89.43\%$ of the variance — which is what the table's other column says.

8. PCA does not delete columns; it replaces them with linear combinations of *all* of them, so all forty tests must still be run on every new patient. On `breast_cancer`, 26 of the 30 PC1 loadings exceed 0.10 and the largest is only 0.2609 — no subset can be dropped without changing the component. What you want is **attribute subset selection**: run RFE or an L1 model with

the two expensive tests excluded from the candidate set, then compare against the full model and decide whether the accuracy difference is worth the cost. That last decision belongs to the clinician, not to the selector.

9. There are three problems and they are not equally serious.

Problem 1, and it is essentially the whole number. Step (iii) chooses features using y on all 60 patients, and step (iv) then cross-validates on those chosen columns. Every held-out fold has already voted on which columns survive. Measured on data with *no signal at all* — 500 noise columns and coin-flip labels, the same shape as this table — that procedure reports 0.8475 ± 0.0293 against an honest 0.5342, an overstatement of +0.3133, on twenty repetitions out of twenty. A reported 0.85 on 500×60 is exactly what you get from nothing.

Problem 2, smaller. Step (ii) fits the scaler on all the rows before any split. Lecture 5 measured this at +0.0034 with a small training set, falling to +0.0002 with a large one. Real, tiny, and free to fix.

Problem 3, not leakage but still wrong. `MinMaxScaler` on long-tailed gene-expression data compresses the bulk of each column towards zero — Section 3.4 measured a single outlier squeezing the good values into $1/71$ of the range. Use `RobustScaler`, or `StandardScaler` after a power transform.

Corrected:

```
pipe = Pipeline([
    ("scale", RobustScaler()),
    ("select", SelectKBest(f_classif, k=20)),
    ("model", LogisticRegression(max_iter=5000)),
])
print(cross_val_score(pipe, X, y, cv=5).mean())
```

Every step is refitted on four fifths of the data inside each fold. If the honest number comes back near 0.5, that is the result: with 60 patients and 500 genes there may be nothing there.

10. A defensible design, with the evidence for each step.

The six numeric columns. Scale them. Four orders of magnitude is the same situation as the age-and-income table, where the wider column supplied 99.995% of the squared distance. Choose the scaler after looking at the histograms: `StandardScaler` if they are reasonably symmetric, `RobustScaler` if they are not, and `MinMaxScaler` only if you have already run an outlier check — one typo cost min–max a factor of 71. If the final model is a tree, scaling changes nothing at all (Section 3.5, +0.0000), so skip it and keep the readable thresholds.

The ordinal column. `OrdinalEncoder(categories=[["bronze", "silver", "gold"]])`, because the default is alphabetical and would give `bronze < gold < silver`. Print `categories_` to check.

The 12-level nominal column. One-hot, with `handle_unknown='ignore'` so an unseen level at prediction time does not raise; twelve columns on 50,000 rows is nothing. Use `drop='first'` if the model is unregularised and anyone will read the coefficients: without it the design matrix has condition number 3.4×10^{15} and the coefficients are not unique.

The 1800-level city column. Do not one-hot it as it stands: our measured case (256 levels, 600 rows, no signal) moved the training R^2 up 0.14 and the test R^2 down 0.12. Keep the commonest few dozen cities and map the rest to `other`; or use a count or target encoding computed *inside* the fold; or replace the column with something meaningful such as city population or region.

Reduction and order. You need a readable model, so prefer selection to PCA: Section 6.5 showed 26 of 30 loadings above 0.10, so no column is dropped and no component has a name. Use an L1 model or RFECV, report the criterion, and be prepared for the answer that selection does not help. Everything then goes into one `ColumnTransformer` inside one `Pipeline`, and the pipeline goes inside the cross-validation. Measured cost of getting that wrong: +0.0006 for a leaked PCA, +0.0034 for a leaked scaler, +**0.3133** for a leaked feature selection.

11 References and further reading

All links were checked on 22 August 2026; the scikit-learn metrics and NumPy `corrcoef` pages were added and checked on 16 September 2026. Free resources are marked [free].

Textbooks

- Han, Kamber and Pei, *Data Mining: Concepts and Techniques*, 3rd edition, Morgan Kaufmann, 2012. **The prescribed text.** §3.4 data reduction — §3.4.1 the overview and the three kinds, §3.4.2 wavelet transforms, §3.4.3 principal components analysis, §3.4.4 attribute subset selection, §3.4.5–3.4.8 numerosity reduction and sampling; §3.5.2 normalisation by min–max, z-score and decimal scaling.
- Müller and Guido, *Introduction to Machine Learning with Python*, O’Reilly, 2016. §3.3 pre-processing and scaling (the four scalers, and the “scale the training and test data the same way” figure); §3.4 dimensionality reduction, with §3.4.1 on PCA; Ch. 4 on representing data — §4.1 categorical variables and one-hot encoding, §4.3 binning, §4.4 interactions and polynomials, §4.5 automatic feature selection (univariate, model-based and iterative). Notebooks [free]: https://github.com/amueller/introduction_to_ml_with_python
- James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*. [free] §3.1.3 assessing the accuracy of a fitted line — the R^2 statistic and why it equals the squared correlation for a single input (the two measures used from Section 2.2 on); §6.3 dimension reduction methods (principal components regression and partial least squares — what to do when PCA’s unsupervised nature is a problem); §12.2 on PCA, the variance-explained curve and the scree plot. <https://www.statlearning.com/>
- Kuhn and Johnson, *Feature Engineering and Selection*. [free] The best single book on this lecture. Ch. 5 encoding categorical predictors, including high cardinality and unseen levels; Ch. 6 engineering numeric predictors; Ch. 10 the filter/wrapper/embedded taxonomy; Ch. 11 greedy search, including RFE. <http://www.feats.engineering/>
- Deisenroth, Faisal and Ong, *Mathematics for Machine Learning*. [free] Ch. 10, “Dimensionality Reduction with PCA”: both derivations — maximum variance and minimum reconstruction error — done properly, with a companion notebook. <https://mml-book.github.io/>
- Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*, 2e, Springer. §7.10.2, “The Wrong and Right Way to Do Cross-validation”, is the canonical statement of the selection bias of Section 7.8; §3.4 ridge and lasso; §14.5 principal components.
- Wes McKinney, *Python for Data Analysis*, 3e. [free] §7.2 on `pd.get_dummies` and `pd.cut/pd.qcut` — the pandas half of Sections 2.4 and 4. <https://wesmckinney.com/book/data-cleaning>

Online courses

- **NPTEL, Data Mining** — Prof. Pabitra Mitra, IIT Kharagpur. The data-preprocessing and data-reduction lectures are the closest match to this material in the vocabulary the

examination uses. [free] <https://nptel.ac.in/courses/106105174>

- **NPTEL, Introduction to Machine Learning** — Prof. Balaraman Ravindran, IIT Madras; the reference course for the semester, with a dimensionality-reduction week covering PCA. [free] <https://nptel.ac.in/courses/106106139>
- **SWAYAM** — the national portal hosting NPTEL. [free] <https://swayam.gov.in/>
- **Kaggle Learn, Feature Engineering** — six short browser lessons (mutual information, creating features, PCA as feature engineering, target encoding); two or three hours, and the closest practical companion to this lecture. [free] <https://www.kaggle.com/learn/feature-engineering>
- **Google Machine Learning Crash Course** — the “Working with numerical data” and “Categorical data” modules. [free] <https://developers.google.com/machine-learning/crash-course>

Videos

- StatQuest with Josh Starmer, *StatQuest: Principal Component Analysis (PCA), Step-by-Step* — the eigen-decomposition of Section 6.2 at a whiteboard, including loadings and the scree plot. Watch this one first if PCA has not clicked. [free] <https://youtu.be/FgakZw6K1QQ>
- StatQuest with Josh Starmer, *StatQuest: PCA main ideas in only 5 minutes!!!* — the short version, for revision. [free] https://youtu.be/HMOI_1kzW08
- Krish Naik, *Standardization Vs Normalization- Feature Scaling* — the distinction of Section 3.3, with the formulas. [free] <https://youtu.be/mnKm3YP56PY>
- CampusX, *One Hot Encoding | Handling Categorical Data | Day 27 | 100 Days of Machine Learning* — one-hot in pandas and scikit-learn, including the dummy variable trap and high cardinality. [free] <https://youtu.be/U5oCv3JKWKA>
- Mahesh Huddar, *Feature Selection | Wrapper | Filter | Embedded Intrinsic Method in Machine Learning* — the taxonomy of Section 7 in examination vocabulary. [free] <https://youtu.be/GJnSBhxWEEg>

Documentation and interactive explainers

- Victor Powell and Lewis Lehe, *Principal Component Analysis explained visually* — drag the points and watch the components move, in 2D, 3D and a 17-dimensional example. The best five minutes you can spend on this lecture. [free] <https://setosa.io/ev/principal-component-analysis/>
- scikit-learn user guide: *Preprocessing data* (§8.3, the scalers and the encoders); *Decomposing signals in components* (§2.5, PCA); *Feature selection* (§1.13, with the filter, RFE, `SelectFromModel` and pipeline sections); *Unsupervised dimensionality reduction* (§8.5); *Pipelines and composite estimators* (§8.1, including `ColumnTransformer`). [free] <https://scikit-learn.org/stable/modules/preprocessing.html>
<https://scikit-learn.org/stable/modules/decomposition.html>
https://scikit-learn.org/stable/modules/feature_selection.html
https://scikit-learn.org/stable/modules/unsupervised_reduction.html
<https://scikit-learn.org/stable/modules/compose.html>

- scikit-learn user guide, *Metrics and scoring* (§3.4) — the R^2 score section gives Equation (2) and its edge cases; and the NumPy reference for `numpy.corrcoef`, which computes Equation (1). [free] https://scikit-learn.org/stable/modules/model_evaluation.html#r2-score
<https://numpy.org/doc/stable/reference/generated/numpy.corrcoef.html>
- scikit-learn, *Common pitfalls and recommended practices* — read the sections on inconsistent preprocessing and on data leakage before you next report a score. [free] https://scikit-learn.org/stable/common_pitfalls.html
- Three scikit-learn examples that reproduce results from these notes: *Compare the effect of different scalers on data with outliers*; *Comparison of F-test and mutual information*; *Recursive feature elimination with cross-validation*. [free] https://scikit-learn.org/stable/auto_examples/preprocessing/plot_all_scaling.html
https://scikit-learn.org/stable/auto_examples/feature_selection/plot_f_test_vs_mi.html
https://scikit-learn.org/stable/auto_examples/feature_selection/plot_rfe_with_cross_validation.html
- MLU-Explain — Amazon’s visual explainers; the bias–variance and train/test/validation pages are the background for Section 7.8. [free] <https://mlu-explain.github.io/>

Sources: Han, Kamber and Pei, *Data Mining: Concepts and Techniques* (3e), Ch. 3.; Müller and Guido, *Introduction to Machine Learning with Python*, Ch. 3–4.; James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*, Ch. 5–6, 12.; Scikit-learn documentation (preprocessing, model selection), accessed 2026-08-04.